Why we pace & the routes

Heart block

After valve, septal or root surgery, TAVR, endocarditis or MI. Often recovers in days.

Slow rhythms

Sinus node dysfunction, slow AF or junctional, drug-related; bridge to a permanent device.

Backup

Rate set below the patient's own: paces only if the native rhythm fails.

⚠ **Pacemaker-dependent** = no reliable escape rhythm (rate under 40 or asystole when pacing stops). Any pacing failure is an **emergency**: pads on, ZOLL ready for transcutaneous pacing, provider aware.

The routes

		TRANSVENOUS (TVP)		
	Epicardial wires	**Right IJ**	**Brachial**	**Femoral**
What it is	Ventricular wires sewn on in the OR, exiting below the incision.	Bipolar lead to the RV apex via the right IJ; tip confirmed on CXR.	Same lead through an arm vein (brachial / basilic).	Same lead through the femoral vein in the groin.
Typical use	Open-heart patients who may need pacing.	Our usual route: heart block, bradycardia, post-TAVR, endocarditis.	Occasional: when the neck can't be used.	Uncommon.
Nursing priority	Label wires; cap unused ends; keep dry; thresholds rise over days; bleeding or tamponade risk at removal.	CXR after placement; dressing intact; check the **cm at the hub**; limit neck movement.	**Highest risk of dislodgement or movement:** tape the wires, limit arm movement, verify the cm at the hub after any activity.	Keep the **leg straight**; **don't bend the patient at the waist**; **check pulses** in that leg.

Our unit standards

1. Temporary pacemakers should be set to **VVI**.
2. Test the **stimulation threshold every shift**.
3. Assess the **underlying rhythm every shift**, unless the MD defers it or the patient is too unstable.
4. **Notify the provider** if the settings need to be changed or if the pacemaker **loses capture despite troubleshooting**.

Document every shift: mode · rate · capture threshold · output · underlying rhythm

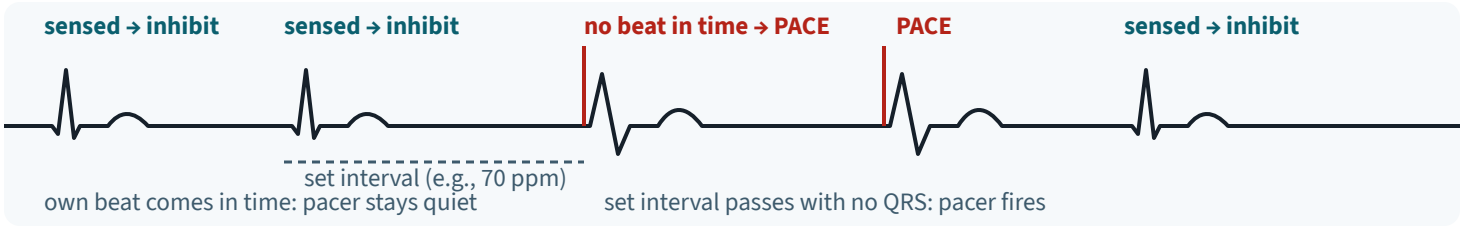
VVI & the sensitivity fence

What VVI means

V **Paces** the **Ventricle**. The wire on or in the ventricle delivers the pacing pulse.

V **Senses** the **Ventricle**. The same wire listens for the patient's own QRS.

I **Inhibits** when it senses a beat: it holds its pulse and **restarts its timer**.



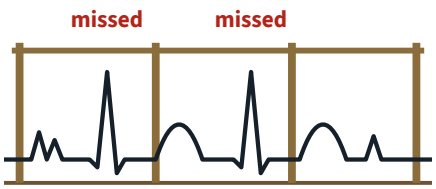
Patient's own rate above the set rate → pacer is inhibited: SENSE light flashes, no spikes.
Own rate below the set rate → pacer fills in: PACE light, spikes, wide paced QRS.

Limits: no atrial kick, so cardiac output can drop in stiff ventricles (AS, LVH, after AVR). **ASYNC (VOO)** turns sensing off: only briefly, with monitoring and the provider aware (R-on-T risk).

Sensitivity = a fence the signal must jump over

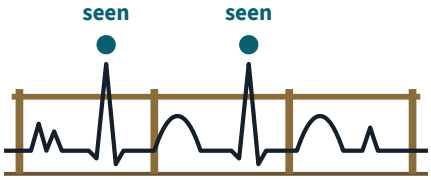
The mV number is the **height of the fence**. The pacer only "sees" signals tall enough to clear it. **Raise the fence** (higher mV) = **less** sensitive. **Lower the fence** (lower mV) = **more** sensitive.

Fence too HIGH



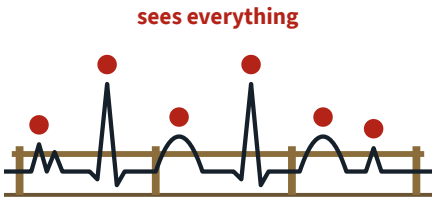
Higher mV. Real QRS can't clear it, so the pacer fires anyway: **undersensing**, spikes on T waves.

Fence JUST RIGHT



Every QRS clears it; T waves and noise don't. Usually **half the height of the QRS or lower**.

Fence too LOW



Lower mV. T waves and noise jump over too, so the pacer holds back: **oversensing**, pauses.

Unit standard: sensitivity is set in the OR / cath lab. If it isn't sensing correctly, **adjust the sensitivity until it senses correctly**, document in the chart, then **notify the MD**.

Temporary pacemaker controls

Rate	Output	Sensitivity	Lock	Lights
30–200 ppm	0.1–25 mA	0.8–20 mV or ASYNC	always on	PACE · SENSE · LOW BATT

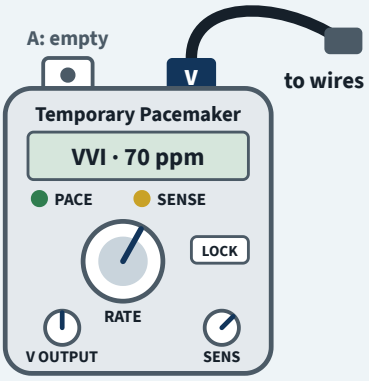
A flashing PACE light only means a pulse was sent. It does **not** prove capture: check the monitor and a pulse.

On arrival & every shift

On arrival to the unit

1. **Patient first:** rhythm, pulse, BP.
2. Ensure the pacemaker is in **VVI** mode and **review the settings** with whoever brought the patient to the unit.
3. **Sensitivity** is set in the OR / cath lab. Not sensing correctly? See page 2.
4. **Trace the connection:** verify the cable is plugged into the pacemaker and the wires are secured.
5. **Battery:** no LOW BATT light; spare AAs at the bedside.
6. **Confirm capture:** spike → paced QRS → pulse.
7. **Transvenous:** note the **cm at the hub**; balloon syringe **locked** (page 4).
8. **Only if the patient is stable:** test the stimulation threshold and assess the underlying rhythm (page 5).
9. **Lock** the controls and **document**.

Hook-up



Only the **right (V) port** is used: it paces the ventricle. The **left (A) port** is for atrial pacing and stays empty, since we always use VVI.

Route-specific care

Epicardial

Verify the wires are **not pulled out** from the patient.

Not in use: **secure, insulate, cap** and keep dry.

Removed when the **chest tubes** come out.

Right IJ

Tip in the RV apex: **CXR** after placement.

Paced QRS has an **LBBB pattern**. A new **RBBB pattern** suggests perforation or LV position.

Check the **cm at the hub** (page 4).

Brachial

Highest risk of dislodgement or movement. Secure the wires with tape and limit movement of that arm.

Verify the **cm at the hub** after any activity (page 4).

Every shift

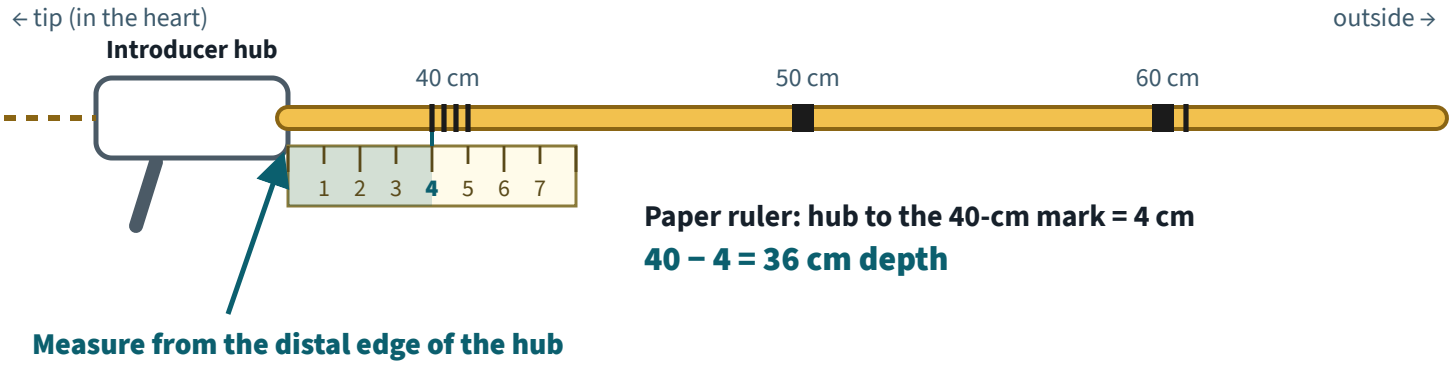
- ☐ Mode **VVI**; settings match the **last assessment or report**; **lock** on.
- ☐ **Stimulation threshold** tested.
- ☐ **Battery** OK; spares at the bedside.
- ☐ **Site:** dressing, exit sites, wires taped and secured.
- ☐ **Capture and sensing** on the monitor, with a pulse.
- ☐ **Underlying rhythm** assessed (unless MD deferred or too unstable).
- ☐ **Connections** tight; cable secured, not hanging.
- ☐ **Transvenous:** cm at the hub unchanged; syringe **locked**.

Document every shift

Mode	Rate	Capture threshold	Output	Underlying rhythm

Transvenous: depth & balloon lock

Reading the depth: measure at the hub



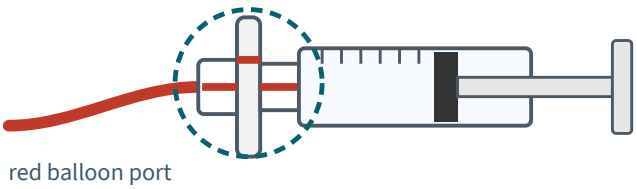


1. **Find the distal edge of the hub**: where the catheter comes out of the introducer.
2. **Find the first mark outside the hub.** Each **thin line = 10 cm**; each **thick band = 50 cm** (a thick band plus 1 thin line = 60 cm).
3. **Measure with a paper ruler** from the distal edge of the hub to that mark, then subtract (40-cm mark, 4 cm out = **36 cm**).
4. **Compare** with the cm from report and your last check. Check every shift and **after any activity**, especially brachial wires.

Pulled out, or pushed in too deep? Don't push it back in or pull it back yourself. Secure it, check capture and **notify the provider**.

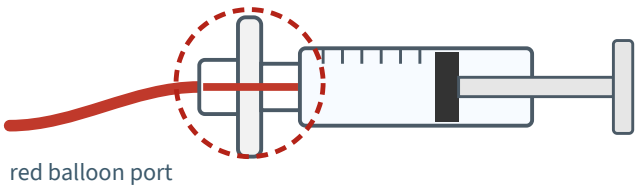
Balloon syringe: make sure it is LOCKED

Locked: correct



Hub and syringe red lines match, but the **lock's red line is NOT straight**.

Unlocked: lock it now



Red line runs **straight across**.

The balloon is only used during placement and stays **deflated**. Keep the syringe attached and the valve **locked**. Check it every shift and after any movement.

Threshold & underlying rhythm

STIMULATION (CAPTURE) THRESHOLD

The lowest output (mA) that still captures every beat. Set the output 2–3 × the stimulation threshold.

Usual thresholds
Transvenous: under 1–2 mA.
Epicardial: 1–5 mA, **rising over days**.
A big jump from last shift? Look for a cause (page 6).

How to test the stimulation threshold

1. **Note** the rate, the rhythm and whether it is capturing.
2. **Make sure every beat is paced.** If the patient's own rate is faster, raise the pacer rate about 10 ppm above it, if tolerated.
3. **Watch** the monitor and the arterial line the whole time.
4. **Turn the output down slowly**, a little at a time.
5. **Stop at loss of capture:** a spike with no QRS after it and no arterial pulse.
6. **Turn it right back up** until 1:1 capture returns.
7. The lowest output with 1:1 capture = the **stimulation threshold**.
8. **Set the output** to 2–3 × the stimulation threshold.
9. **Return the rate** to its previous setting and **lock**.
10. **Document** the threshold and the output.



Pacemaker-dependent? Loss of capture means **no heartbeat**. Go down in small steps and turn the output straight back up the moment capture is lost.

How to assess the underlying rhythm

Every shift, unless the **MD defers it** or the patient is **too unstable**.

1. **Watch** the monitor and arterial line the whole time.
2. **Lower the rate slowly** (about 10 ppm at a time). Don't just turn the pacer off.
3. When the pacer stops firing (SENSE light, no spikes), **identify the patient's own rhythm and rate** and save a strip.
4. **Stop right away** and return to the previous rate if no rhythm appears, the BP drops or the patient has symptoms.
5. **Restore** the previous rate and **lock**.
6. **Document** the rhythm and rate, or "no underlying rhythm at __ ppm".

Example note: VVI · rate 70 · capture threshold 2 mA · output 6 mA · underlying rhythm: complete heart block, ventricular escape at 34.

Recognize & troubleshoot

Patient first. Unstable: adjust the settings (box below). No perfusing rhythm: **ZOLL pads, transcutaneous pacing**, ACLS. Then check **in this order**:

1 · Battery | **2 · Connections** | **3 · Output** | **4 · Sensitivity** | **5 · Lead / patient**

Failure to capture

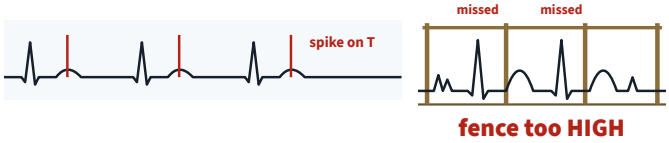


Looks like: spikes not followed by a paced QRS; no pulse with that beat.

Causes: rising threshold (K^+, acidosis, hypoxia, ischemia, drugs), lead moved, loose or broken wire, low battery, perforation.

Do: raise output (to max if needed) · **turn the patient onto their left side** · check battery and every connection · K^+, ABG, CXR.

Undersensing



Looks like: pacer fires as if native beats aren't there; spikes on the QRS or T wave.

Causes: mV number set **too high** = not sensitive enough (fence too high); lead moved, small QRS, loose connection. **Risk:** R-on-T → VT/VF.

Do: turn the mV number DOWN (more sensitive) until it senses correctly · confirm VVI · check the lead and connections · document, then notify the MD.

Oversensing

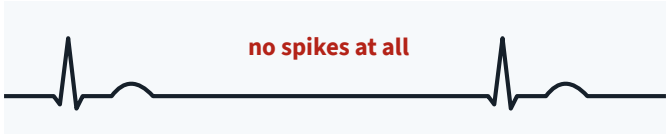


Looks like: pauses; no spike when one is due; SENSE light flashes without a real beat.

Causes: mV number set **too low** = too sensitive (fence too low); T waves, shivering, electrical interference, loose connection.

Do: remove the cause · **turn the mV number UP** (less sensitive) until it senses correctly · document, then notify the MD.

Failure to pace (no output)



Looks like: no spikes and no PACE light while the rate is below the set rate.

Causes: dead battery, pacer off, cable disconnected, device failure (or severe oversensing).

Do: check power and every connection · change batteries · **swap in a new pacemaker** · TCP if no rhythm.

Patient unstable?

It is OK to **adjust the pacer settings as needed** (rate, output, sensitivity) to get a perfusing rhythm.

Once you find settings that work, **notify the MD**.

Notify the provider (order set)

- Loss of capture
- Sudden increase in capture threshold
- Failure to sense despite troubleshooting
- Signs of hemodynamic instability with temp pacing
- Bleeding at the insertion site

Complications & safety

Perforation or tamponade: think of it early

Signs: falling BP, rising CVP, narrowing pulse pressure, muffled heart sounds, chest pain, new rub, **sudden loss of capture**, paced QRS turning to an **RBBB pattern**, **hiccups** or diaphragm twitching with each paced beat.

Do: notify the provider now · urgent echo · pads on, TCP ready · fluids and drips per order · prepare for pericardiocentesis or return to the OR.

Other complications

Complication	Look for	Do
Dislodgement / migration	Loss of capture or sensing, new QRS shape, cm at the hub changed.	Raise output, turn onto the left side, notify; don't push it back in. Limit neck or arm movement.
Ventricular ectopy / VT	PVCs or VT from the lead tip; spikes on T waves.	Check sensing, CXR, K^+ and Mg^{2+}. VT: **pulse or no pulse?** ACLS.
Diaphragm stimulation	Hiccups or abdominal twitching in time with pacing.	Notify; assess for perforation.
Pneumothorax / arterial puncture	Dyspnea, falling SpO_2, one-sided absent breath sounds; neck hematoma.	CXR; pressure to bleeding; notify.
Infection	Redness, drainage, warmth at the site; fever.	Site care per policy; cultures per order.
Phlebitis / vein thrombosis	Arm or neck swelling, pain, color change.	Hourly limb checks; notify; ultrasound per order.
Bleeding / tamponade after wire removal	Low BP, fast HR, rising CVP, chest pain after the wires come out.	Coags and anticoagulant timing per policy first; bed rest, frequent vitals; treat as tamponade.

Safety every time

Microshock: gloves when touching wires; cap or insulate bare ends; keep dry; grounded equipment only.

Secure and label generator, cable and wires.

Never delay defibrillation for the pacemaker; re-check capture and threshold afterward.

Transport: pacer secured, spare batteries, ZOLL with pads.

References

UpToDate (uptodate.com):

Temporary cardiac pacing

Pacing system malfunction: evaluation and management

Modes of cardiac pacing: nomenclature and selection